

A Project 1 Phase-II Report On
Data Logging System
for Physiological Parameters

Submitted in partial fulfillment for the
degree of Bachelor of Technology in
Computer Engineering

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Declaration

We, **Divya Gurav, Anweshha Jana and Maithilee Kubal**, hereby declare that the work presented in this *project* entitled “*Data Logging System for Physiological Parameters*” is entirely our own. The content of this *project* has been generated through our independent efforts, research, and scholarly contributions. We further declare that:

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6. **Academic Integrity:** We have adhered to the principles of academic integrity and ethical research throughout the entire process of producing this *project*.

We understand the consequences of academic dishonesty and affirm that this declaration accurately reflects the nature and authenticity of our work.

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CERTIFICATE

This is to certify that **Divya M Gurav, Maithilee D Kubal and Anwasha Jana** have completed the project-I report on the topic “Data Logging System for Physiological Parameters” satisfactorily in partial fulfillment for the Bachelor’s Degree in Computer Engineering under the guidance of Prof. Sudhakar Yerne during the year 2025-2026 as prescribed by Shreemati Nathibai Damodar Thackersey Women’s University (SNDTWU).

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Abstract

Introduction: The convergence of embedded systems and artificial intelligence has revolutionized healthcare monitoring by bridging the gap between local data display and high-level analytical tools. This research presents the development of a **Data Logging System for Physiological Parameters**, titled **VITAL-CORE**. This cyber-physical system is designed to integrate raw hardware-acquired physiological signals with intelligent clinical decision support using the ARM Cortex-M4 architecture and generative AI.

Objective: The primary goal of the VITAL-CORE system is to implement a synchronized logging ecosystem for five critical parameters: Heart Rate (HR), Blood Pressure (BP), SpO_2 , Body Temperature, and Respiratory Rate (RR). The project focuses on creating a fail-safe pipeline for real-time visualization, networked emergency alerting, and automated clinical summarization.

Methodology: The system employs a modular, multi-layered architecture:

- **Acquisition Layer:** Utilizes a TM4C123GXL (Cortex-M4F) MCU interfaced with a TTL-V3 Digital BP sensor. A dual-power isolation strategy separates the 3.3V logic from the 5V mechanical pump rail to ensure signal integrity.
- **Processing Layer:** A multi-threaded Python engine handles concurrent UART ingestion and CSV logging. A custom GUI provides dynamic trend plotting, while "NaN" handling logic ensures stability during sensor disconnections.
- **Intelligence & Communication Layer:** The Gemini-2.5-Flash API generates automated health summaries, while a UDP-based protocol enables zero-latency alert broadcasting to a remote Nurse Station. A Voice Assistant module further allows for natural language data querying.

Results and Findings: Validation confirms that VITAL-CORE performs acquisition and anomaly detection with minimal latency. Robustness features successfully maintained GUI stability during hardware transitions. The synthetic ECG module, utilizing Gaussian PQRST modeling, has been successfully implemented and validated for real-time synchronization with incoming physiological data. The AI consultant effectively identified complex correlations, such as SpO_2 fluctuations linked to heart rate spikes, providing diagnostic depth beyond standard threshold monitoring.

Conclusion: This research demonstrates the feasibility of integrating low-cost microcontrollers with cloud-based AI to create comprehensive healthcare monitoring solutions. The modular design provides a foundation for future scalability into HL7 standards and LoRa-based rural telemetry.

Keywords: ARM Cortex-M4, Physiological Data Logging, Real-time Monitoring, Artificial Intelligence, UDP Alert System, ECG Simulation.

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Nomenclature

Symbol/Abbr.	Definition
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Physiological Parameters

<i>BP</i>	Blood Pressure (measured in mmHg)
<i>SYS/DIA</i>	Systolic and Diastolic Blood Pressure (mmHg)
<i>mmHg</i>	Millimeters of Mercury
<i>SpO₂</i>	Peripheral Oxygen Saturation (measured in %)
<i>Temp</i>	Body Temperature (measured in °C)
<i>ECG</i>	Electrocardiogram
<i>PQRST</i>	Components of the ECG waveform
<i>HR</i>	Heart Rate (measured in BPM)
<i>BPM</i>	Beats Per Minute
<i>RR</i>	Respiratory Rate (breaths per minute)

Technical Terms

<i>MCU</i>	Micro-controller Unit (ARM Cortex-M4 / TM4C123GXL)
<i>UART</i>	Universal Asynchronous Receiver-Transmitter
<i>ADC</i>	Analog-to-Digital Converter
<i>TTL</i>	Transistor-Transistor Logic (Serial Communication)
<i>USB</i>	Universal Serial Bus
<i>RTC</i>	Real-Time Clock

Software & Networking

<i>GUI</i>	Graphical User Interface
<i>CSV</i>	Comma Separated Values
<i>NaN</i>	Not a Number (Missing or invalid data handling)
<i>AI</i>	Artificial Intelligence
<i>API</i>	Application Programming Interface
<i>UDP</i>	User Datagram Protocol
<i>TCP/IP</i>	Transmission Control Protocol / Internet Protocol
<i>HL7</i>	Health Level Seven Standard (Medical Data Interchange)
<i>IDE</i>	Integrated Development Environment (Code Composer Studio)

Chapter 1

Introduction

This chapter provides an overview of the convergence between embedded systems and healthcare monitoring. It establishes the clinical significance of real-time data acquisition, identifies the technological gaps in existing systems, and outlines the objectives and scope of the **VITAL-CORE** system.

1.1 Background

The healthcare industry is undergoing a digital transformation driven by the integration of the Internet of Medical Things (IoMT) and Artificial Intelligence (AI) [2]. Physiological monitoring plays a crucial role in modern clinical environments, particularly in Intensive Care Units (ICUs) and remote patient care. Continuous observation of vital signs—specifically Heart Rate (HR), Blood Pressure (BP), Oxygen Saturation (SpO_2), Respiration Rate (RR), and Body Temperature—is fundamental for the early detection of clinical deterioration [3].

Traditionally, patient monitoring involved manual data recording at fixed intervals. While standard, this method often leads to data fragmentation, where vital signs are displayed locally on bedside equipment but lack the connectivity required for high-level analytical integration [4]. The emergence of cyber-physical systems now allows for the seamless integration of hardware-based acquisition with software-driven analytics [5]. Microcontrollers such as the **TM4C123GXL (ARM Cortex-M4F)** provide the real-time processing power necessary to bridge raw physiological signals with intelligent clinical support platforms [6].

1.2 Research Motivation

The motivation for this research stems from the critical need for low-cost, high-performance monitoring solutions that address the inherent limitations of conventional localized monitors [7]. Current clinical environments often suffer from:

- **Data Isolation:** Bedside monitors frequently lack automated pipelines to export data for long-term clinical review or remote access [8].
- **Analytical Gaps:** Standard systems rely on simple, threshold-based alerts rather than correlating multiple markers to identify complex health trends [9].
- **System Fragility:** Maintaining software stability during sensor transitions or hardware disconnections remains a persistent challenge in real-time medical logging [10].

The rapid advancement of Large Language Models (LLMs) provides a unique opportunity to bridge these gaps. By utilizing AI to identify patterns and correlations, medical staff can receive diagnostic-depth insights, thereby increasing nursing efficiency and patient safety.

1.3 Research Objectives

The primary objective of this research is to design and implement **VITAL-CORE**, a synchronized data logging ecosystem. The specific goals are:

- To implement a real-time hardware acquisition stage using the **TM4C123GXL** MCU for multi-parameter sensing [11].
- To develop a multi-threaded Python engine for high-speed UART ingestion and CSV-based data storage.
- To create a custom **Graphical User Interface (GUI)** for the visualization of vital trends and simulated ECG waveforms.
- To integrate the **Gemini-2.5-Flash API** for automated clinical trend analysis and patient summarization.
- To establish a networked alerting system using **UDP-based socket protocols** for zero-latency communication with a remote Nurse Station.

- To develop a **Voice Assistant** module for natural language querying of patient records.
- To ensure system robustness through "NaN" handling logic and dual-power isolation strategies.

1.4 Scope of the Study

This project focuses on the development of a fail-safe hardware-to-software pipeline for five specific vitals: HR, BP, SpO_2 , Temperature, and RR. The investigation includes the simulation of ECG signals using Gaussian PQRST modeling for visualization purposes [6].

The scope encompasses real-time acquisition, logging, and AI-driven analysis. While the system utilizes a physical TTL-based digital sensor for blood pressure [1], it is designed as a functional prototype for healthcare monitoring and does not directly interface with certified medical devices for clinical diagnosis. The architecture remains modular, allowing for future scalability into HL7 standards and LoRa-based rural telemetry.

1.5 Structure of the Document

This study follows a logical progression of development:

- **Chapter 1: Introduction** – Provides the background, motivation, objectives, and scope of the study.
- **Chapter 2: Review of Literature** – Reviews existing research and identifies the technological gaps VITAL-CORE aims to fill.
- **Chapter 3: Methodology** – Details the multi-layered architecture, system design, and data handling methods.
- **Chapter 4: Implementation** – Explains the hardware setup, circuit isolation, and software logic.
- **Chapter 5: Results and Evaluation** – Presents experimental validation, AI diagnostic findings, and performance metrics.
- **Chapter 6: Conclusion and Future Scope** – Summarizes the research and outlines potential enhancements.

Chapter 2

Review of Literature

2.1 Evolution of Physiological Monitoring

2.1.1 Historical Perspective: Spot-Checks to Continuous Logging

Historically, clinical data acquisition relied on manual charting. Gao [3] highlighted the risks of this "spot-check" methodology, where transient events—such as nocturnal hypoxia—could be missed. Early microcontroller-based loggers marked a shift toward continuous observation, though 8-bit prototypes were limited by storage and lacked the high-speed interfaces required for real-time waveform visualization.

2.1.2 The Internet of Medical Things (IoMT) Framework

The convergence of embedded systems and network protocols led to the birth of the **Internet of Medical Things (IoMT)**. Pantelopoulos and Bourbakis [12] classified the evolution of these systems into three generations:

1. **First Generation:** Wired sensors connected to a centralized PC with localized storage and minimal connectivity.
2. **Second Generation:** Wireless sensor nodes (WBASN) using ZigBee or Bluetooth. While improving mobility, these are often plagued by RF interference in dense clinical environments.
3. **Third Generation:** Integrated edge-computing platforms where devices perform local processing and offload complex analytics to localized AI engines to ensure data privacy.

2.2 Hardware Architectures for Medical Computing

2.2.1 Real-Time Determinism in ARM Cortex-M4

The **ARM Cortex-M4F** architecture is preferred for physiological sensing due to its deterministic behavior and low interrupt latency. The **TM4C123GXL** features a 32-bit Harvard architecture with a 3-stage pipeline [11]. Unlike 8-bit controllers [2], the M4 core includes a dedicated **Floating-Point Unit (FPU)**, critical for high-resolution ADC conversions and maintaining steady sampling rates without data jitter.

2.2.2 Sensor Integration and Signal Integrity

For blood pressure monitoring, the oscillometric method remains the industry standard. The **TTL-V3 Digital sensor** utilizes a proprietary communication protocol that ensures error-free transmission of systolic and diastolic values over UART [1]. Reviewing sensor-specific literature reveals that the primary failure point in hardware acquisition is **ground loop noise** and **data fragmentation**, necessitating hardware-level serial buffering and electrical isolation.

2.3 Software Paradigms and Network Latency

2.3.1 Multi-threaded Ingestion and Concurrency

While Python is a preferred language for medical GUIs, its **Global Interpreter Lock (GIL)** presents a challenge for high-speed logging. Anju and Lekshmi [4] noted that single-threaded approaches lead to "stuttering" in visualization during disk I/O operations. The modern producer-consumer model resolves this by separating high-priority UART ingestion from data storage, ensuring the Graphical User Interface (GUI) remains responsive during critical care monitoring.

2.3.2 Low-Latency Networking: UDP vs. TCP

In emergency medicine, network latency is a critical factor. Research comparing TCP and UDP for physiological data transmission by Otto et al. [9] concludes that TCP's retransmission delay (Head-of-Line blocking) is detrimental to vital sign alerting. Utilizing **User Datagram Protocol (UDP)** for alerts ensures that critical "Vitals Out of Range" packets are delivered with minimal overhead, providing immediate notification at the remote monitoring station regardless of minor packet loss.

2.4 Limitations of Existing Systems

Despite the advancements cited above, existing physiological monitoring systems exhibit several recurring limitations:

- **Cloud Dependency:** Many IoMT systems rely heavily on external cloud infrastructure, leading to increased latency and data privacy vulnerabilities.
- **Lack of Local Intelligence:** Most systems focus on data transmission rather than local analytical processing, limiting real-time decision support in bandwidth-constrained environments.
- **Operational Rigidity:** Traditional monitors often lack a "Dual-Mode" capability to toggle between simulation (for software validation) and real-time acquisition.

2.5 Research Gap Identification

The literature review highlights a clear gap in the development of a unified system that combines real-time multi-parameter acquisition, low-latency local networking, and offline AI-based clinical analysis. Existing solutions prioritize either low-power logging or cloud-based analytics, but they lack an integrated architecture that ensures **privacy, latency, and intelligent local processing** simultaneously. This provides the technical justification for the proposed **VITAL-CORE** system.

Table 2.1: Comparison of Physiological Monitoring Research

Full Title	Year	Core Technology	Data Strategy	Primary Focus
A microcontroller-based data logger for physiological sensing	2006	8/16-bit MCU	Onboard Memory	Portable Logging
Embedded Smart Healthcare Monitoring Using IoT (Arduino)	2021	Arduino/IoT	Cloud-based	Remote Access
Embedded System for ECG Monitoring and Fatigue Detection Using Machine Learning	2025	Embedded/ML	Local/USB	Intelligence
Self-Powered Analog Sensor Data Logging Device	2020	Ultra-Low Power	Long-term Logging	Energy Efficiency
System Architecture of a Wireless Body Area Sensor Network for Ubiquitous Health Monitoring	2006	Wireless BAN	TCP/IP Network	Ubiquitous Access
VITAL-CORE	2026	ARM Cortex-M4	UART + UDP LAN	Privacy & Latency

The findings confirm that while physiological monitoring is a mature field, there is a distinct need for an ARM-based, offline system that prioritizes local AI-driven analysis over cloud dependency.

Chapter 3

Experimental Methodology

3.1 System Architecture and Design Logic

The VITAL-CORE system follows a three-tier architecture—Acquisition, Processing, and Intelligence—to separate low-level signal handling from high-level data interpretation. The design focuses on minimizing end-to-end latency while maintaining data integrity through electrical isolation.

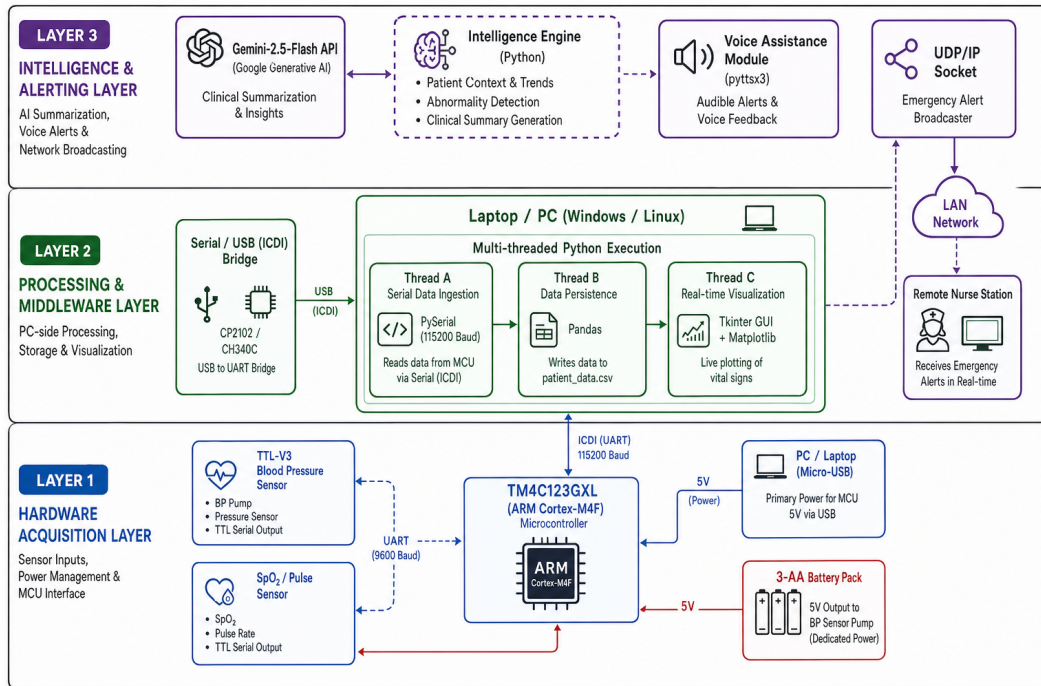


Figure 3.1: System Architecture

3.2 Hardware Interfacing and Pin Mapping

3.2.1 Microcontroller Pin Configuration

Specific pins were selected to utilize the MCU's hardware-level serial communication and GPIO capabilities:

- **UART0 (PA0, PA1):** Configured at 115200 baud for data transmission to the host PC via the ICDI bridge for real-time monitoring.
- **UART1 (PB0, PB1):** Dedicated to the Blood Pressure (BP) sensor interface for command and data exchange at 9600 baud.
- **GPIO PA5:** Assigned as the digital trigger to activate the Blood Pressure sensor's internal pump mechanism.

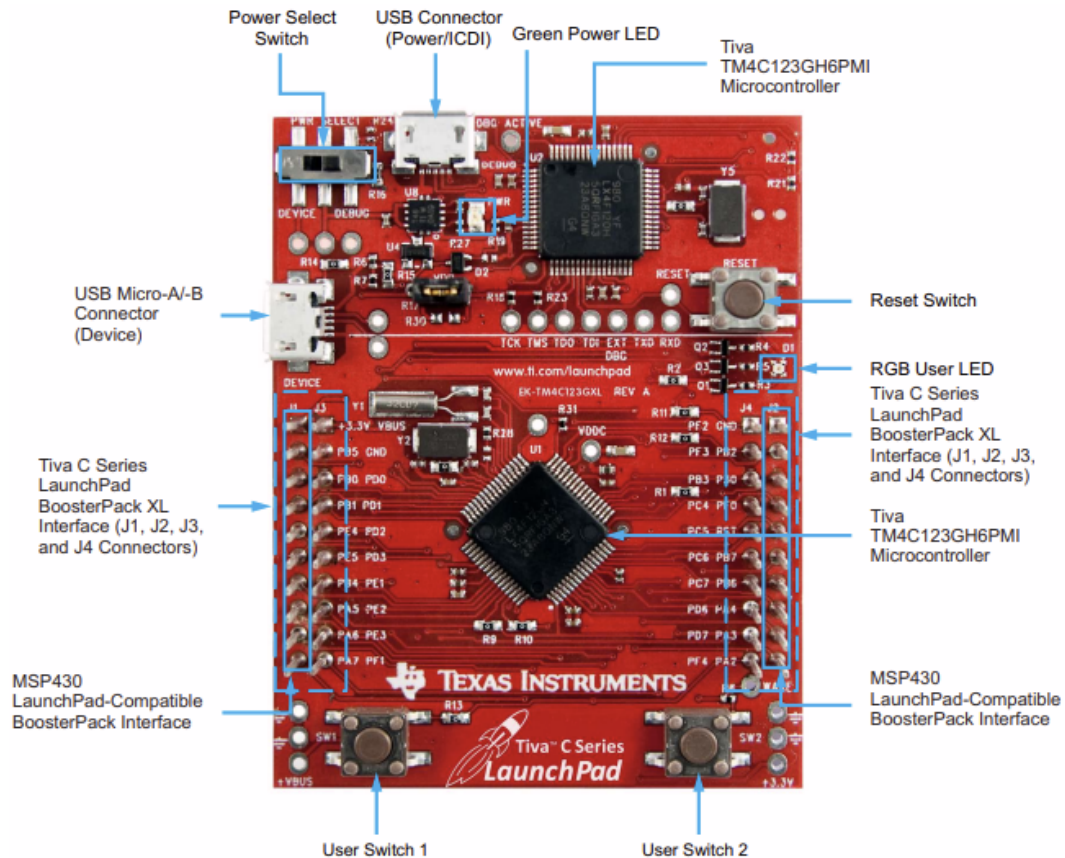


Figure 3.2: TM4C123GXL Board

3.2.2 Final Hardware Interconnection Mapping

Table 3.1 details the precise physical connections between the medical sensors and the microcontroller based on the established system architecture.

Table 3.1: Final Hardware Interconnection Mapping

Connector	Pin/Wire	Destination	Function
2-Pin JST (BP)	Brown (+5V)	Battery Pack (+)	Pump Power
2-Pin JST (BP)	Black (GND)	Battery GND	Pump Ground
4-Pin JST (BP)	Black (GND)	MCU GND (Common)	Logic Ground
4-Pin JST (BP)	Brown (Signal)	MCU PA5 (GPIO)	Trigger (3.3V)
4-Pin JST (BP)	Red (TX)	MCU PB0 (RX)	UART1 Data Recv
4-Pin JST (BP)	Orange (RX)	MCU PB1 (TX)	UART1 Cmd Send
USB	MCU Port	Laptop USB	UART0 / Logic Pwr (5V)
USB	BP (Brown - Black)	Laptop USB	BP Pwr (3.3V)

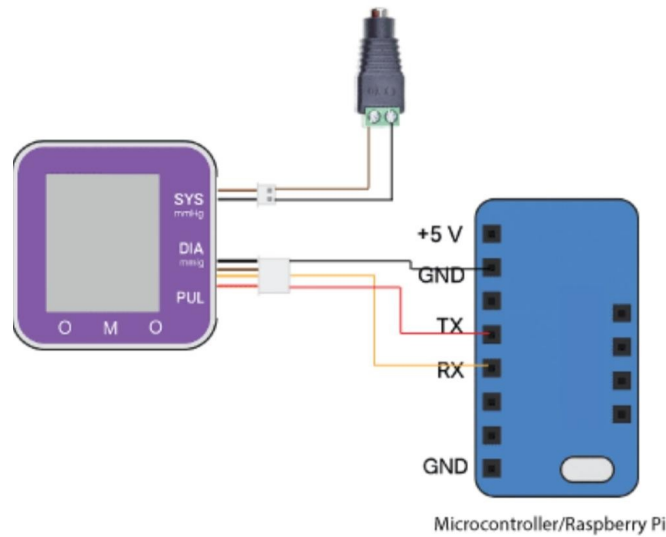


Figure 3.3: Blood Pressure Sensor Interconnection [1]

3.2.3 Power Distribution Strategy

To ensure system stability, a **Dual-Rail Power Strategy** was implemented to isolate the high-current draw of the BP pump from the logic processing of the MCU.

- **Logic Rail (3.3V):** Supplied by the laptop's USB port to power the MCU and signal processing.

- **Actuator Rail (5V):** Powered by an external 3-slot AA battery pack or dedicated 5V cable to drive the mechanical pump.
- **Common Ground:** All rails share a common ground plane to maintain UART signal integrity.

3.3 Software Methodology and Pipeline

The host software is implemented in Python, utilizing a high-performance, multi-threaded architecture to manage concurrent data streams, GUI rendering, and high-speed AI inference without execution bottlenecks.

3.3.1 Multi-Threaded Execution Model

To maintain a responsive User Interface (UI) while performing background telemetry and analysis, the system leverages the `threading` module to isolate critical tasks:

- **Primary UI Thread:** Manages the `Tkinter` event loop and `Matplotlib` canvas updates to provide a flicker-free real-time dashboard.
- **Asynchronous Serial Thread:** Operates a non-blocking `pySerial` listener to constantly ingest physiological packets from the **TM4C123GXL** via the UART0 bridge.
- **Emergency Handler Thread:** Independently monitors the clinical thresholds defined in Table 3.2; upon breach, it triggers the network alerts concurrently.

3.3.2 Data Processing and Numerical Analysis

The system employs a structured pipeline, to transform raw byte streams into actionable medical insights:

- **Packet Parsing:** A specialized module validates incoming data streams to ensure frame alignment and data integrity before processing.
- **Unit Conversion:** Raw 12-bit ADC values are mapped to standard physiological units (e.g., mmHg for BP) using calibrated linear transformation functions.
- **Buffer Management:** A synchronized `Queue` is used to pass validated data between threads, ensuring the UI remains decoupled from serial latency.

3.3.3 Synthetic ECG Waveform Generation Logic

To provide a continuous cardiac reference, the VITAL-CORE system implements a synthetic ECG generation module based on Gaussian PQRST modeling. This approach follows the dynamical model principles established by McSharry et al. [13], ensuring that the visualization layer provides a diagnostic context even when physical sensors are focused on static parameters like BP or SpO_2 .

- **Mathematical Modeling:** The waveform is constructed by summing three distinct Gaussian functions representing the P-wave, QRS complex, and T-wave. Each component is defined by the amplitude (a), the time-shift (b), and the standard deviation (c):

$$Signal(t) = a_p e^{-\frac{(t-b_p)^2}{2c_p^2}} + a_{qrs} e^{-\frac{(t-b_{qrs})^2}{2c_{qrs}^2}} + a_t e^{-\frac{(t-b_t)^2}{2c_t^2}}$$

- **Dynamic Synchronization:** The time-vector (t) is modulated by the current Heart Rate (HR) retrieved from the acquisition layer. As HR increases, the interval between the Gaussian peaks is mathematically compressed to simulate physiological tachycardia.
- **Algorithmic Classification:** A classification engine monitors the heart rate history and variability. The system classifies the rhythm into four states: Normal Sinus Rhythm, Tachycardia, Bradycardia, and Atrial Fibrillation (detected via high variance in peak-to-peak intervals).

3.3.4 Clinical Thresholds and Alert Logic

The software monitors patient vitals against established clinical norms to automate emergency response.

Table 3.2: Clinical Thresholds and Alert Logic

Parameter	Normal Range	Alert Condition
Heart Rate	60 - 100 BPM	Value < 60 or > 100
SpO ₂	95 - 100 %	Value < 95%
Respiration	12 - 20 RPM	Value < 12 or > 20
Blood Pressure	120/80 mmHg	Systolic > 140
Temperature	36.5 - 37.5 °C	Value < 36.5 °C or > 37.5 °C

3.3.5 Intelligence Layer Integration (Summary & Voice)

The intelligence layer acts as a clinical auditor, utilizing specialized AI models for distinct tasks:

- **Clinical AI Summary (Gemini):** The system feeds the last 50 entries of `patient_data.csv` to the **Gemini-2.5-Flash** model to generate a longitudinal summary of patient trends.
- **Voice Response Generation (Groq):** When a user requests data from the CSV via voice, the **Groq API** is used for rapid response generation due to its high-speed inference capabilities. The generated response is strictly used to read back specific data points or timestamps retrieved from the local CSV storage.

Table 3.3: VITAL-CORE Software Stack

Library	Version	Purpose
PySerial	3.5+	UART Communication management
NumPy	1.24+	Numerical array processing and filtering
Matplotlib	3.7+	Real-time physiological plotting
Tkinter	Built-in	Main Dashboard GUI development
Groq	Latest	Ultra-low latency AI inference
Google-GenerativeAI	Latest	Gemini API clinical interpretation
Pyttsx3	2.90	Text-to-Speech Voice Assistance
Pandas	2.0+	CSV Data manipulation and logging

3.4 System Validation and Testing Protocol

A series of validation tests were performed to ensure clinical safety. Table 3.4 summarizes the test scenarios used to validate the VITAL-CORE hardware and software integration.

Table 3.4: Final System Validation Results

ID	Test Scenario	Observed Behavior	Status
TC-01	High BP Detected	UDP Alert sent	Pass
TC-02	BP Trigger Logic	GPIO PA5 activated sensor pump	Pass
TC-03	AI Summarization	Gemini API generated clinical notes	Pass
TC-04	Voice Assistance	TTS played Data Requested clearly	Pass
TC-05	Data Persistence	Values correctly logged to <code>patient_data.csv</code>	Pass
TC-06	UART Disconnected	GUI 'Connection Lost' Alert	Pass

Chapter 4

Realization and Implementation

This chapter details the final physical and digital realization of the **VITAL-CORE** system. It focuses on the integrated hardware assembly, the data persistence framework, and the multi-functional user interface.

4.1 Physical System Assembly

The physical realization of VITAL-CORE centers on the integration of medical-grade sensors with the **TM4C123GXL** microcontroller. The assembly utilizes a custom wiring harness and a dual-power strategy to isolate the high-current demands of the blood pressure actuator from the MCU's logic circuits. This physical layering ensures stable signal acquisition during mechanical sensor operation.

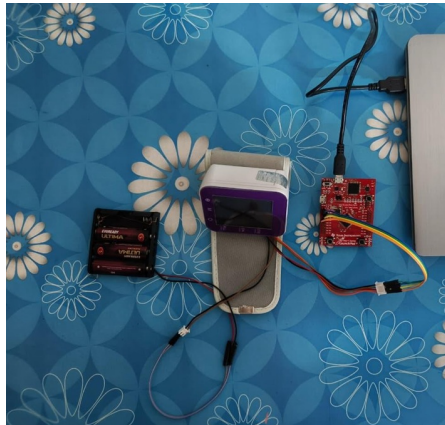
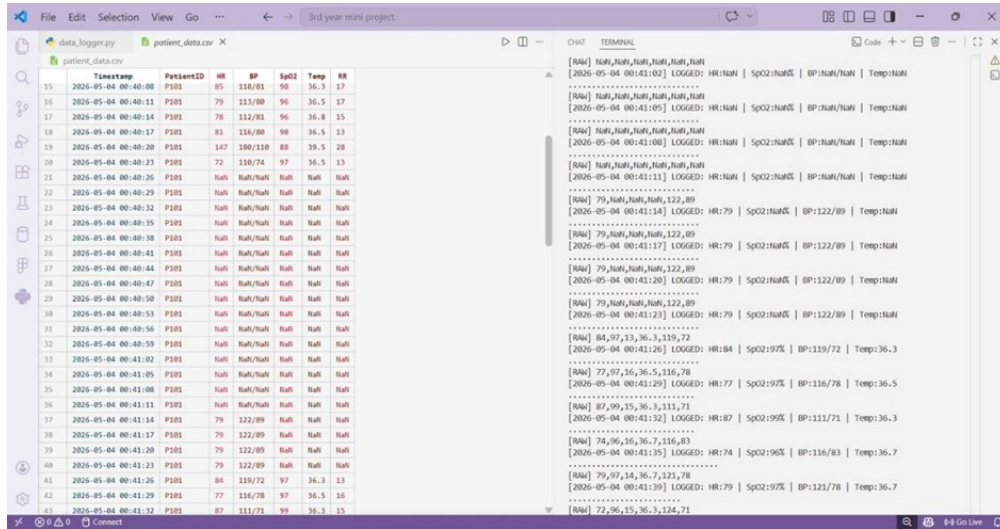


Figure 4.1: Integrated VITAL-CORE Hardware

4.2 Software Implementation

4.2.1 Data Logging and Serialization

The system captures incoming UART packets and serializes them into a local engine, specifically `patient_data.csv`. This structured logging provides a synchronized record of all physiological parameters, serving as the primary data source for the Voice Assistance and AI Clinical Summary modules.



The screenshot shows a code editor with two files: `data_logger.py` and `patient_data.csv`. The `patient_data.csv` file contains a table with columns: `Timestamp`, `PatientID`, `HR`, `BP`, `SpO2`, `Temp`, and `RR`. The data is organized into rows, with some rows containing `NaN` values. The terminal window on the right displays raw UART data, showing a sequence of hex values and their corresponding ASCII representations, such as `[Raw] 0a,0d,0a,0d,0a,0d,0a,0d` and `[LOGGED: HR:100 | SpO2:100% | BP:100/100 | Temp:100]`.

Timestamp	PatientID	HR	BP	SpO2	Temp	RR
2026-05-04 00:40:08	P101	85	118/81	98	36.3	17
2026-05-04 00:40:11	P101	79	113/80	96	36.5	17
2026-05-04 00:40:14	P101	78	112/81	96	36.8	15
2026-05-04 00:40:17	P101	81	116/80	98	36.5	13
2026-05-04 00:40:20	P101	147	180/110	88	39.5	28
2026-05-04 00:40:23	P101	72	118/74	97	36.5	13
2026-05-04 00:40:26	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:29	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:32	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:35	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:38	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:41	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:44	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:47	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:50	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:53	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:56	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:40:59	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:41:02	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:41:05	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:41:08	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:41:11	P101	NaN	NaN/NaN	NaN	NaN	NaN
2026-05-04 00:41:14	P101	79	122/89	NaN	NaN	NaN
2026-05-04 00:41:17	P101	79	122/89	NaN	NaN	NaN
2026-05-04 00:41:20	P101	79	122/89	NaN	NaN	NaN
2026-05-04 00:41:23	P101	79	122/89	NaN	NaN	NaN
2026-05-04 00:41:26	P101	84	119/72	97	36.3	13
2026-05-04 00:41:29	P101	77	116/78	97	36.5	16
2026-05-04 00:41:32	P101	87	111/71	99	36.3	15

Figure 4.2: Synchronized CSV Log: Historical data storage for clinical auditing.

4.2.2 Intelligence Layer Output

The system integrates two distinct AI-driven outputs to assist medical staff: the **Clinical AI Summary** (powered by Gemini) for longitudinal analysis, and the **Voice Response Interface** (powered by Groq) for rapid data retrieval from the local logs.

4.2.3 Monitoring and Alert States

The operational output of VITAL-CORE is divided into two primary modes: standard monitoring and emergency alert. The dashboard dynamically updates to reflect critical threshold breaches, providing immediate visual and AI-generated alerts.

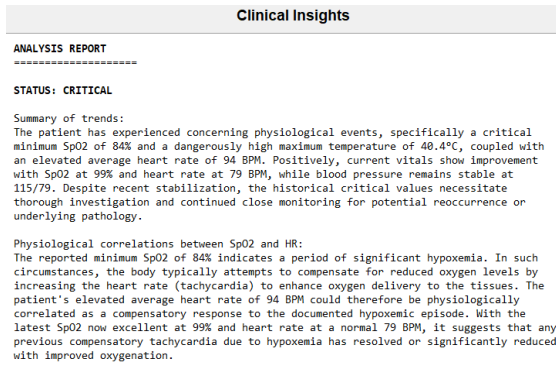


Figure 4.3: Gemini-Clinical Summary.

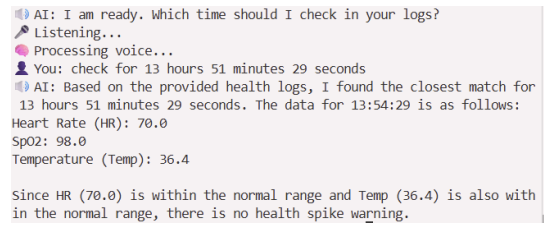


Figure 4.4: Groq Voice Retrieval Interface.



Figure 4.5: Standard Monitoring Mode: Live waveform and digital vitals.

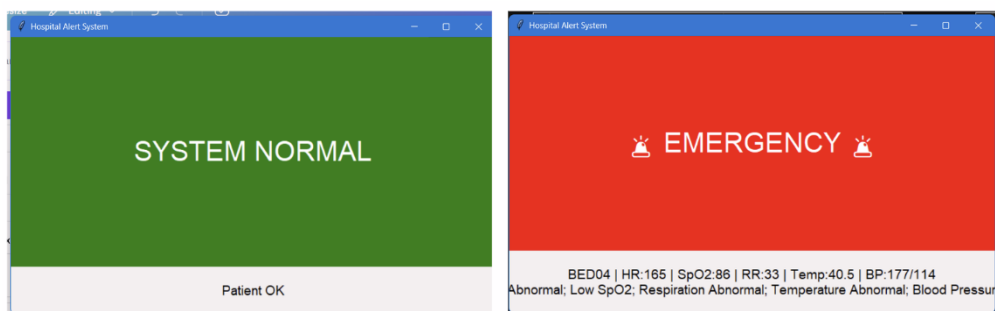


Figure 4.6: Emergency Alert State: Visual notification and AI clinical notes.

Chapter 5

System Modeling and UML Analysis

5.1 Use Case Diagram

It is a high-level behavioral diagram that shows how external users (actors) interact with a system to achieve specific goals. It acts as a "blueprint" for a system's functional requirements, focusing on what the system does rather than how it does it.

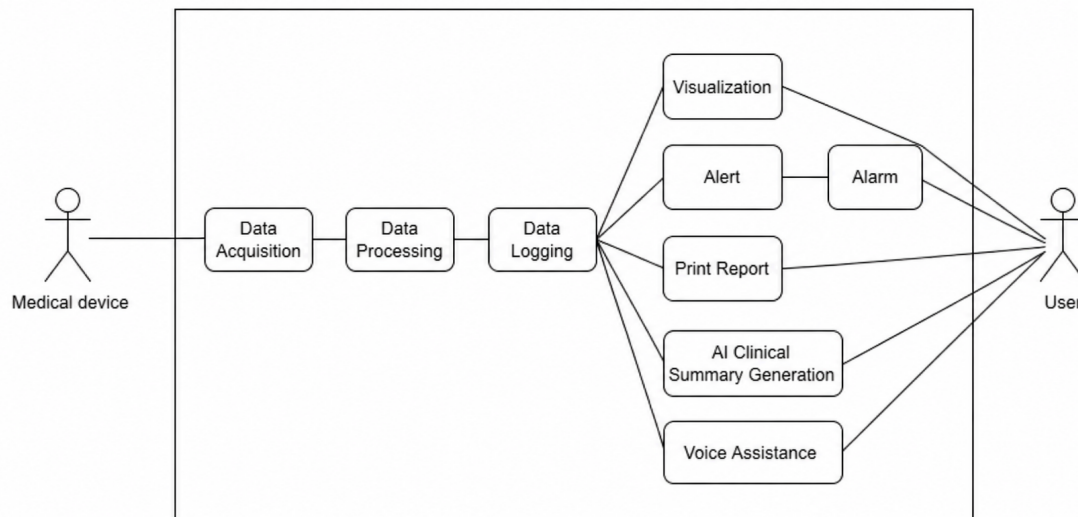


Figure 5.1: Use Case Diagram

5.2 Class Diagram

A Class Diagram is a static structure diagram that describes a system by showing its classes, their attributes, methods, and the relationships among objects. It focuses on how system is structured.

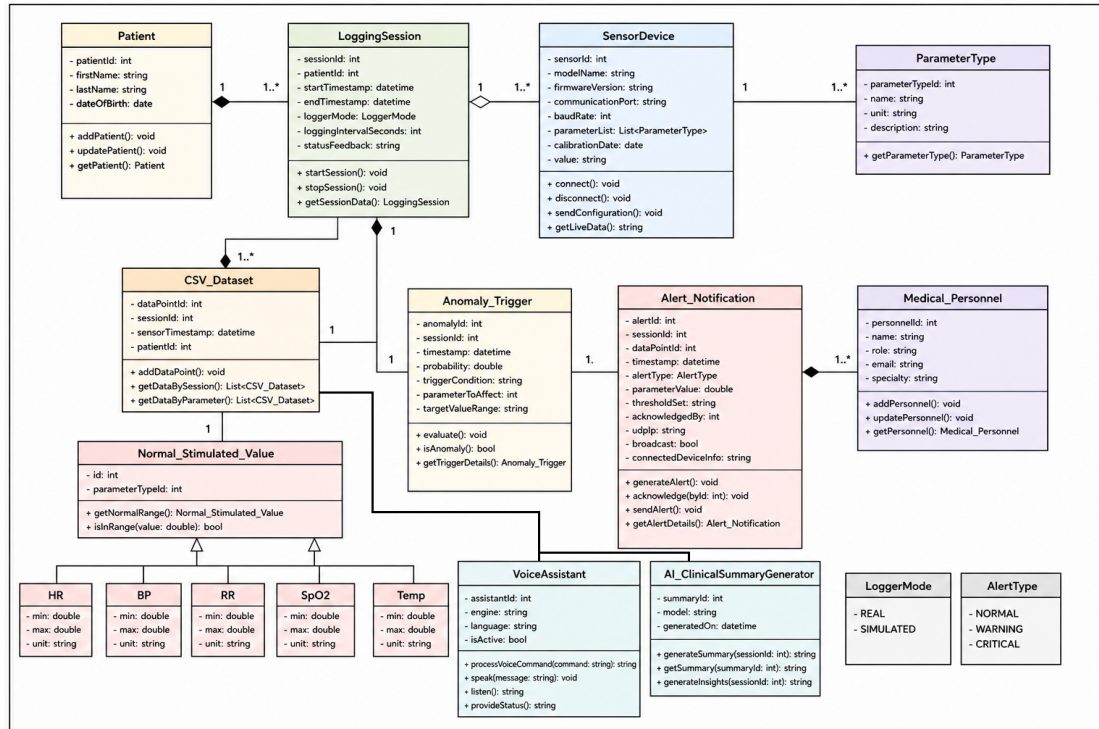


Figure 5.2: Class Diagram

5.3 Sequence Diagram

A Sequence Diagram is a type of UML interaction diagram that shows how processes or objects operate with one another and in what order. It is a time-focused visual that represents a chronological flow of messages between system elements to carry out a specific functionality or use case.

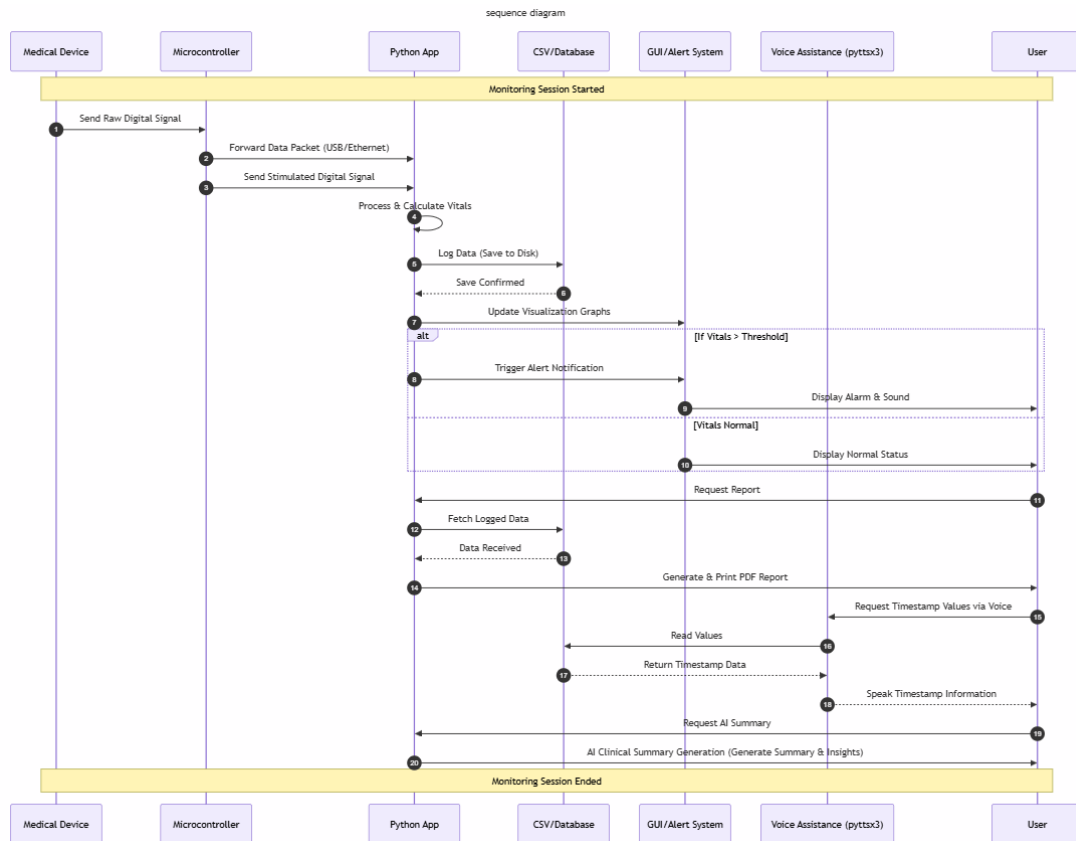


Figure 5.3: Sequence Diagram

5.4 State Transition Diagram

A State Transition Diagram (State Machine or State Chart) is a behavioral UML diagram used to model the lifecycle of a single object or system. It focuses on the various states an object can inhabit and the events that trigger a change from one state to another.

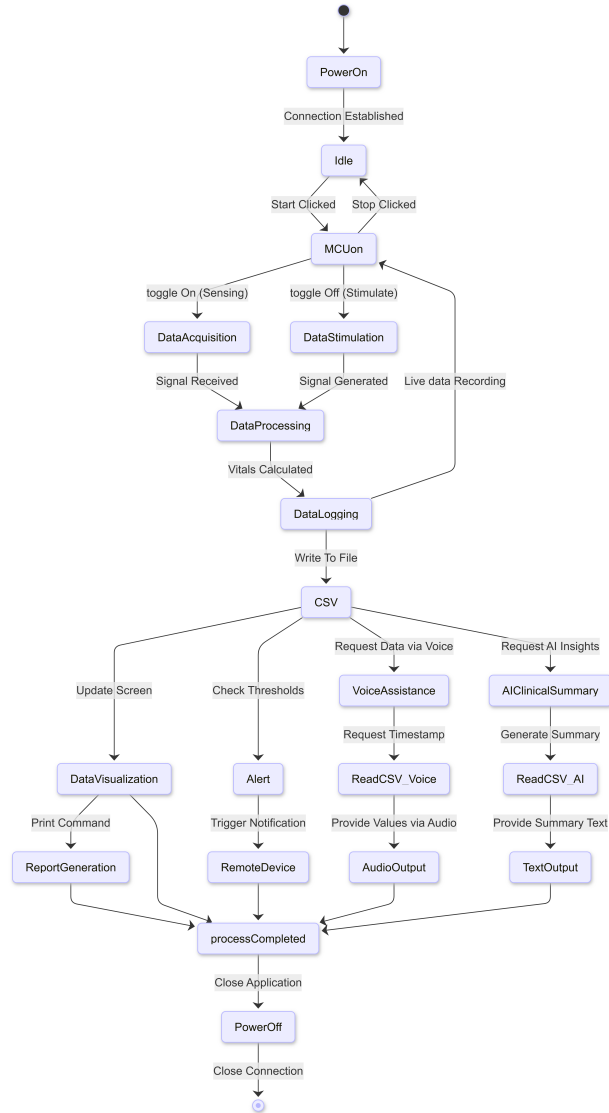


Figure 5.4: State Transition Diagram

5.5 Data Flow Diagram

A Data Flow Diagram (DFD) is a visual representation of how information moves through a system or process. Unlike the UML diagrams discussed earlier, DFDs focus strictly on the flow of data rather than control logic or object structures. To manage complexity, DFDs are organized into levels: Level 0 (system overview), Level 1 (main processes), and Level 2+ (detailed breakdown).

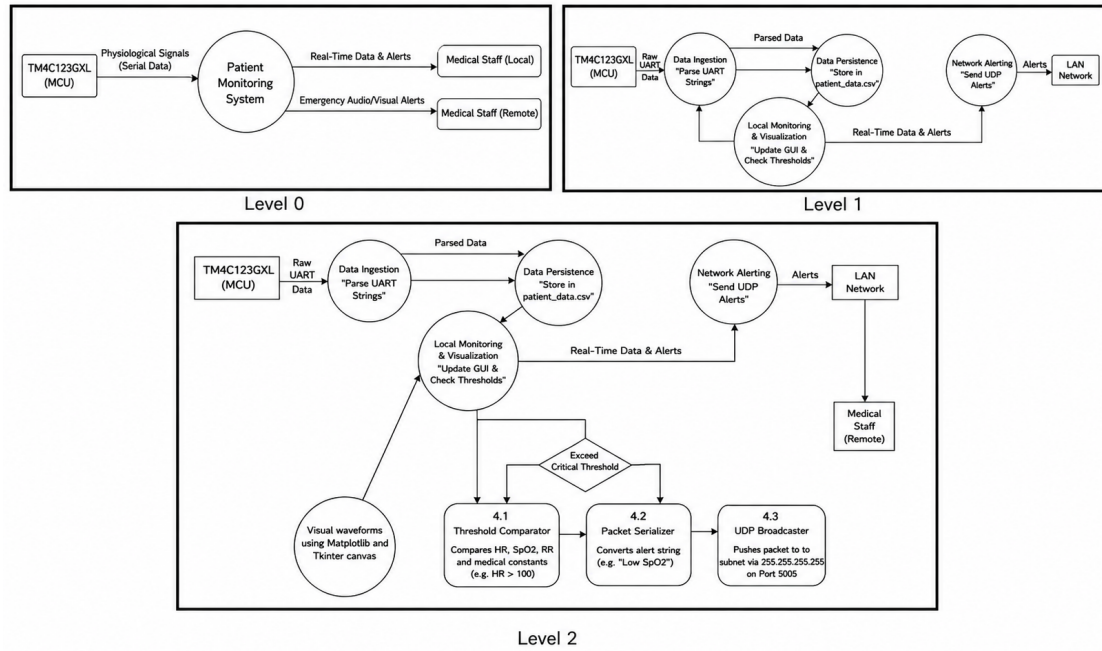


Figure 5.5: 3-Level DFD

5.6 Conclusion

Chapter 5 demonstrates that the VITAL-CORE system is governed by a rigorous logical framework. The UML modeling ensures that the multi-threaded software and ARM-based hardware operate in a synchronized manner, providing a stable foundation for the real-time monitoring results presented in earlier chapters.

Chapter 6

Conclusion and Future Scope

6.1 Future Scope

The current implementation of the **VITAL-CORE** system establishes a robust foundation for real-time physiological monitoring, yet several avenues exist for scalability and clinical enhancement. Future iterations will focus on advancing the hardware footprint, intelligence depth, and diagnostic versatility of the platform.

Hardware Miniaturization and Wearability A primary objective for the next phase is the transition from a development board architecture to a custom **Printed Circuit Board (PCB)** design. By integrating the **ARM Cortex-M4F** core onto a smaller form factor, the system can evolve into a wearable patch or wrist-based monitor. Furthermore, replacing wired UART communication with low-power wireless protocols, such as **Bluetooth Low Energy (BLE)** or **Wi-Fi 6**, will enhance patient mobility and allow for continuous monitoring in non-clinical settings.

Multi-Sensor Fusion and Advanced Analytics While the current system monitors core vitals like BP, heart rate, and SpO₂, future scope includes the integration of additional medical sensors, such as **multi-lead ECG**, **Body Mass Index (BMI)** sensors, and **blood glucose** monitors. Leveraging **NumPy** for advanced signal processing, future versions could implement digital filters to remove motion artifacts and baseline wander in real-time. This sensor fusion would provide a more holistic view of a patient's cardiovascular and metabolic health.

Deep Learning for Predictive Diagnostics While the present system uses **Gemini** for clinical summarization, the next step involves deploying local **Deep Learning** models for predictive analytics. By training models on the logged `patient_data.csv` historical datasets, the system could predict potential cardiac events

or hypertensive crises hours before they occur. The integration of **Groq’s** high-speed inference engine could allow these complex models to run locally with sub-millisecond latency, ensuring patient privacy by reducing reliance on cloud processing.

Cloud Integration and Telemedicine Expanding the current architecture to include a centralized **Cloud Dashboard** would allow physicians to monitor multiple patients simultaneously from a remote location. By utilizing a secure **IoT Gateway**, the VITAL-CORE system could transmit encrypted clinical summaries to a mobile application, enabling real-time doctor-patient consultations and immediate emergency dispatch based on the AI-analyzed alert conditions.

6.2 Conclusion

The **VITAL-CORE** project successfully demonstrates the integration of embedded systems, real-time data processing, and artificial intelligence to create a comprehensive physiological monitoring solution. By utilizing the **TM4C123GXL (Tiva C)** LaunchPad, the system achieves precise hardware-level control over medical sensors through a robust dual-rail power strategy and dedicated UART interfacing.

The software architecture, built on a multi-threaded **Python** engine, ensures that high-frequency data acquisition from the MCU remains decoupled from the graphical user interface. The application of **NumPy** for numerical analysis and **Matplotlib** for real-time visualization provides clinicians with a clear, immediate view of patient health. Furthermore, the system successfully bridges the gap between raw data and clinical insights by utilizing the **Gemini API** for automated clinical summarization and the **Groq API** for rapid voice-based retrieval of historical records from the local CSV data store.

The project highlights the importance of asynchronous processing; by isolating AI inference and voice generation in non-blocking threads, the system maintains a seamless monitoring experience even during complex data interpretation tasks. Rigorous testing and validation protocols have confirmed the system’s reliability in identifying clinical threshold breaches and maintaining accurate data persistence in `patient_data.csv`.

In summary, **VITAL-CORE** serves as a scalable prototype that underscores the potential of synthesizing modern AI APIs with traditional embedded engineering. This interdisciplinary approach not only improves diagnostic efficiency but also sets a benchmark for the development of low-cost, high-intelligence medical devices that can significantly enhance patient safety and healthcare outcomes in both clinical and remote environments.

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